

THE STRUCTURE OF MATHEMATICAL REALITY

Phase IV — Chapter 13

Measurement: How the World Gets Numbers

Note Template

Self-Study Curriculum

How to use this template

Complete these notes **after** the reading and the lecture. The goal is not to copy phrases from the text — it is to reconstruct the ideas in your own language. Leave every item blank on first pass; fill it in from memory, then check.

1. “A measurement scale is a _____.”

The empirical relational structure is:

The numerical relational structure is:

What the map must preserve:

2. The four scale types, each with an example from my own data

Nominal - preserves: _____ admissible transf.: _____

Ordinal - preserves: _____ admissible transf.: _____

Interval - preserves: _____ admissible transf.: _____

Ratio - preserves: _____ admissible transf.: _____

3. Admissible transformations form a _____ (connect to Ch 7).

4. Meaningfulness = invariance

A statistic is meaningful when:
(connect this to Noether / Chapter 5)

5. Averaging, re-examined

Mean is meaningful for:
Median is meaningful for:
Ratios are meaningful for:
So “averaging fails” precisely when:

6. The Velleman-Wilkinson corrective

Why scale type is NOT a rulebook:

7. Three quantities I measure, and the scale type of each

a.

b.

c.

For each, one operation I routinely perform that its scale
does, or does NOT, actually license:
